

# KKBox Churn & Revenue Risk Analytics

## Business Report

**Live Tableau dashboard:** [https://public.tableau.com/app/profile/tianyi.ren4423/viz/Book1\\_17777346522100/D1\\_Overview](https://public.tableau.com/app/profile/tianyi.ren4423/viz/Book1_17777346522100/D1_Overview)

## Executive Summary

This project turns KKBox subscription data into a practical retention analytics workflow. The output is not only a churn model. It is a business decision system that shows which users are at risk, why the risk matters, how much revenue is exposed, and which retention actions should be tested.

The workflow covers the full analytics lifecycle:

1. Build a local analytical warehouse in DuckDB.
2. Transform raw member, transaction, listening, and churn-label data through dbt.
3. Create business-ready marts for monthly KPIs, churn features, model outputs, segment risk, and monitoring.
4. Run statistical tests and churn modelling in Python.
5. Score users by churn probability and assign risk bands.
6. Quantify revenue at risk and retention ROI sensitivity.
7. Deliver Tableau dashboards and a final business report for stakeholder review.

Key results from the current reproducible export:

| Area                               | Result                                  |
|------------------------------------|-----------------------------------------|
| Scored users                       | 970,960                                 |
| Observed churn rate                | 9.0%                                    |
| Calibrated LightGBM AUC            | 0.805                                   |
| Top-decile churn recall            | 39.7%                                   |
| Top-decile lift                    | 3.97x                                   |
| Risk-adjusted revenue exposure     | 4.64M price units                       |
| Critical + high-risk concentration | 1.47% of users / 71.5% of risk exposure |

The main business message is clear: retention budget should not be spread evenly across the user base. The critical and high-risk groups are small enough to target but large enough in revenue exposure to justify action. However, discounting should be tested carefully because a high-risk score does not automatically mean a discount is profitable.

## End-to-End Architecture

The project is structured like an enterprise analytics pipeline. Raw CSV files are not used directly in dashboards. They are first modelled into governed tables, then converted into decision-ready outputs.

What this shows:

- **Raw data:** KKBox members, transactions, user logs, and churn labels.
- **DuckDB warehouse:** local analytical database used to process large CSV data efficiently.
- **dbt marts:** repeatable SQL transformations with staging, intermediate, core, and analytics layers.
- **Python modelling:** statistical tests, logistic regression baseline, LightGBM model, calibration, risk scoring, and monitoring.
- **Tableau dashboards:** stakeholder-facing views for KPIs, churn drivers, model performance, revenue risk, actions, and monitoring.
- **Business action:** risk-band targeting, A/B test design, and ROI-based retention decisions.

## Data Foundation and dbt Lineage

The data model follows a layered structure:

## End-to-End Analytics Architecture

From raw subscription data to governed marts, predictive scoring, dashboard communication, and retention decisions.

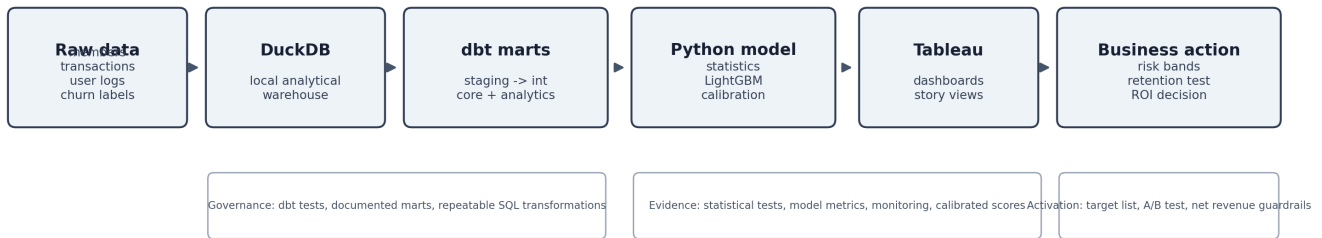


Figure 1: End-to-end analytics architecture

| Layer           | What it does                                     | Examples                                                                                                              |
|-----------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Staging         | Clean names, cast types, standardise raw columns | <code>stg_members</code> , <code>stg_transactions</code> , <code>stg_user_logs</code> , <code>stg_churn_labels</code> |
| Intermediate    | Create reusable user-level summaries             | member profile, transaction summary, engagement windows                                                               |
| Core marts      | Build business facts and dimensions              | <code>dim_user</code> , subscription transactions, listening daily, churn labels                                      |
| Analytics marts | Feed modelling and dashboards                    | churn features, segment summary, monthly KPIs, risk scores                                                            |

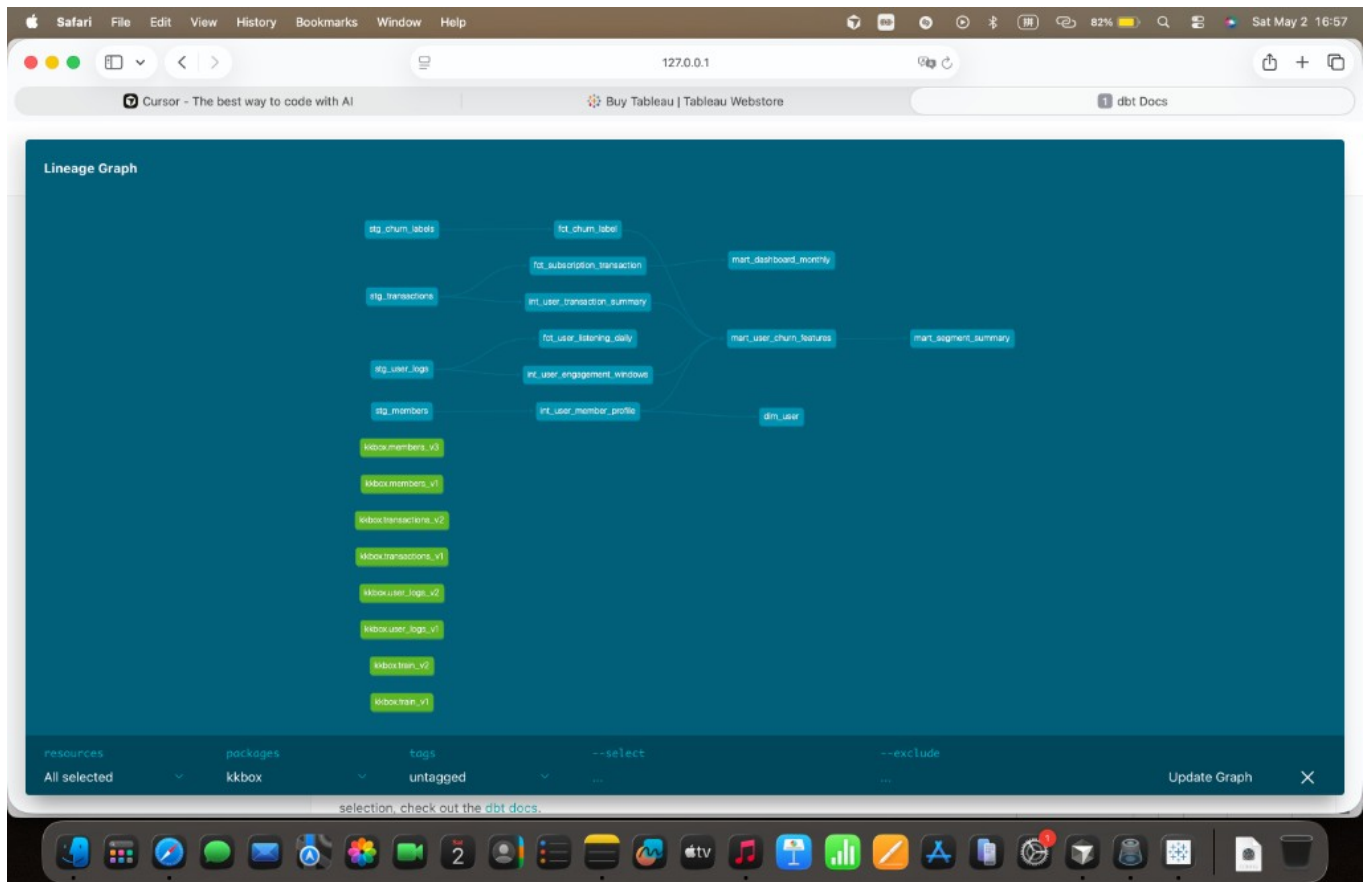


Figure 2: dbt lineage graph

The lineage graph demonstrates that the project has a governed data foundation. The final dashboards and model

- churn labels,
- modelling features,
- dashboard-ready KPI tables,
- user-level risk score outputs.

## 2. dbt Transformation Layer

I used dbt to make the SQL workflow repeatable and easier to audit. Instead of writing one-off SQL queries, the project uses named models and clear layer boundaries.

This matters because a real analyst often needs to answer:

- Where did this metric come from?
- Which source tables feed this dashboard?
- Can the transformation be rerun safely?
- Are model and dashboard definitions consistent?

## 3. Statistical Evidence

The statistical layer checks whether major churn patterns are supported by evidence.

| Test                            | Finding                                 | Business use                                                             |
|---------------------------------|-----------------------------------------|--------------------------------------------------------------------------|
| Auto-renew off vs auto-renew on | 67.8% churn vs 13.5% churn, $p < 0.001$ | Auto-renewal status should be a core retention signal.                   |
| Discount band vs churn          | $p < 0.001$                             | Discount policy is related to churn and should be evaluated financially. |
| Registration channel vs churn   | $p < 0.001$ , Cramer's V around 0.20    | Acquisition channel can inform lifecycle targeting.                      |

This section is intentionally focused on the tests with clean, reproducible outputs. Weak or incomplete test outputs are not used as headline claims.

## Churn Modelling

The modelling workflow compares an interpretable baseline with a stronger boosted model.

| Model               | AUC   | Log loss | Brier score | Top-decile recall | Top-decile lift |
|---------------------|-------|----------|-------------|-------------------|-----------------|
| Logistic Regression | 0.765 | 0.566    | 0.189       | 35.6%             | 3.56x           |
| LightGBM raw        | 0.804 | 0.288    | 0.078       | 39.8%             | 3.98x           |
| LightGBM calibrated | 0.805 | 0.239    | 0.066       | 39.7%             | 3.97x           |

The calibrated LightGBM model is the final reporting model because it provides both ranking power and usable probabilities. The top-decile lift is especially useful for business teams: if the team can only act on a limited list, the model finds churners far more efficiently than random targeting.

## Revenue Risk and Segment Prioritisation

The model output is converted into a business metric:

**risk-adjusted revenue exposure = churn probability x expected revenue**

This lets the project answer a more useful question than “who might churn?”: which users and segments put the most revenue at risk?

Risk-band summary:

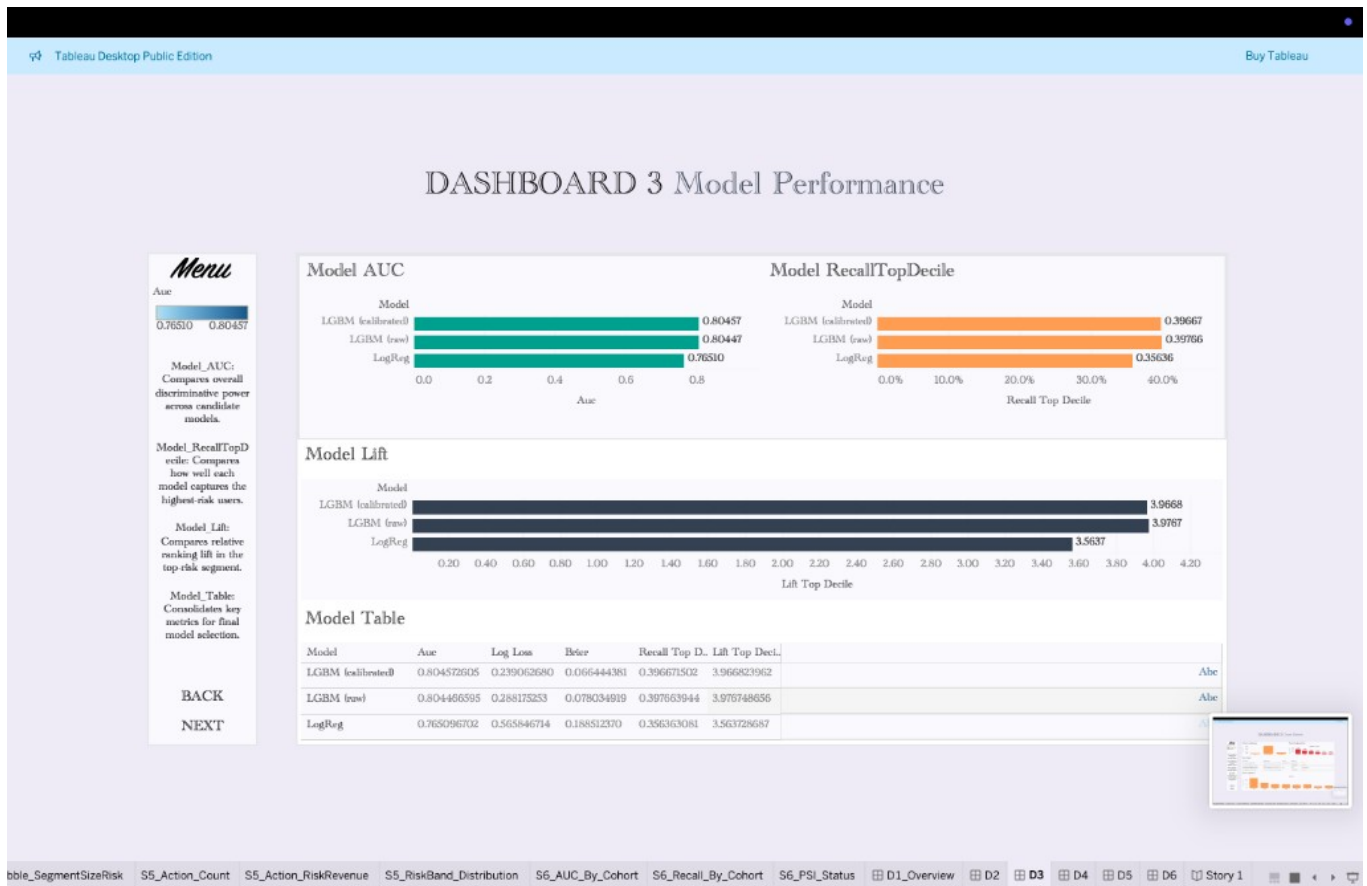


Figure 3: Model performance dashboard

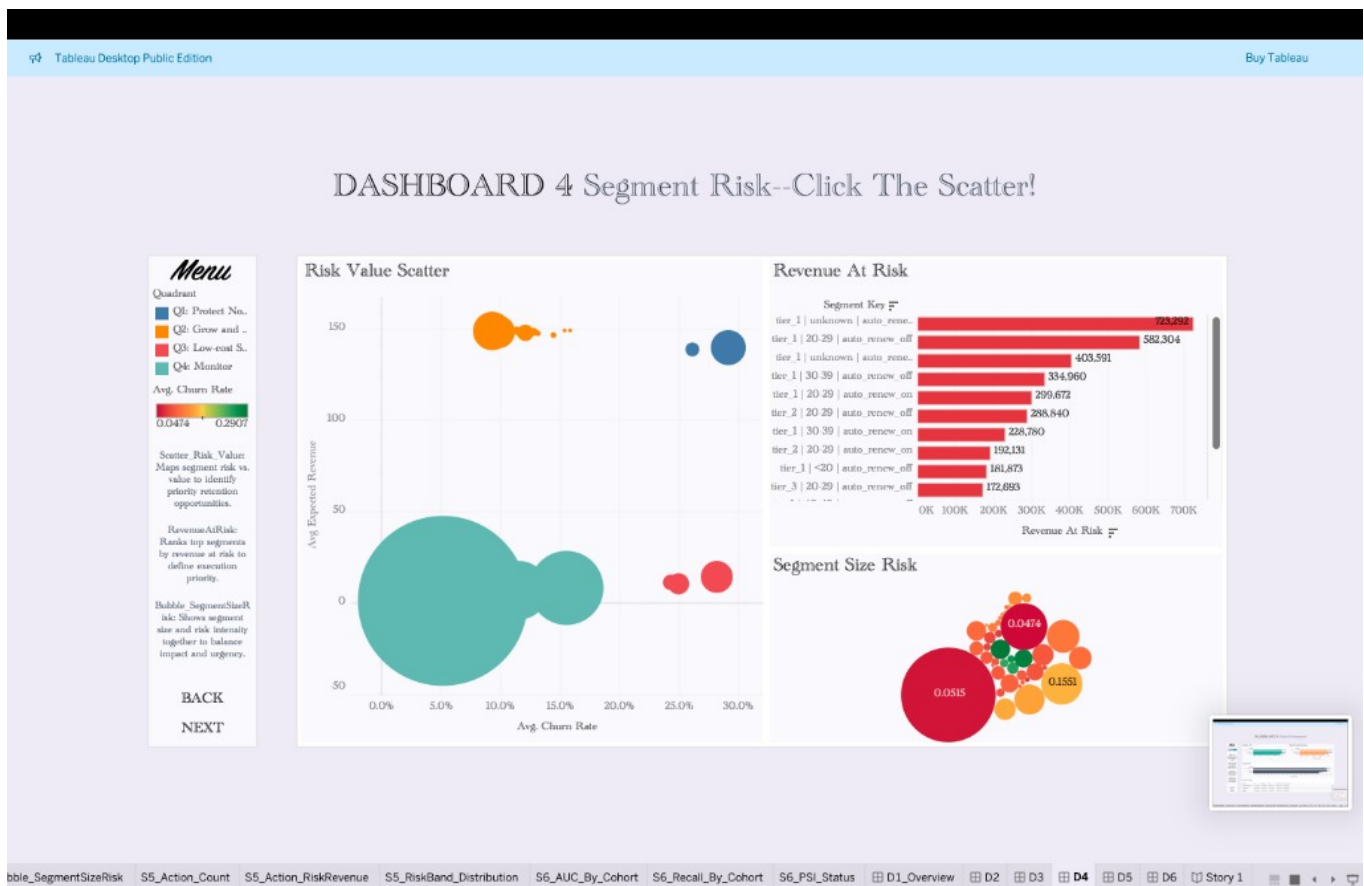


Figure 4: Segment risk dashboard

| Risk band | Users   | Average churn probability | Expected revenue | Risk-adjusted exposure | Recommended action                            |
|-----------|---------|---------------------------|------------------|------------------------|-----------------------------------------------|
| Critical  | 9,812   | 92.2%                     | 3.04M            | 2.89M                  | Personalised offer + customer-service contact |
| High      | 4,438   | 70.5%                     | 0.62M            | 0.43M                  | Targeted renewal discount test                |
| Medium    | 215,515 | 40.3%                     | 1.20M            | 0.47M                  | Low-cost re-engagement push                   |
| Low       | 741,195 | 16.9%                     | 13.96M           | 0.85M                  | No active paid intervention                   |

The critical and high-risk groups are only 1.47% of scored users but account for 71.5% of risk-adjusted exposure. That is the core prioritisation insight.

## Revenue Impact and ROI Sensitivity

Revenue at risk is not the same as financial impact. In a real business, a retention campaign should be judged by incremental value:

$\text{expected saved revenue} = \text{targeted expected revenue} \times \text{churn reduction}$

$\text{discount cost} = \text{discount rate} \times \text{expected revenue} \times \text{expected renewal probability after treatment}$

$\text{net revenue impact} = \text{expected saved revenue} - \text{discount cost}$

$\text{ROI} = \text{net revenue impact} / \text{discount cost}$

The sensitivity table below targets critical + high-risk users: 14,250 users and 3.67M price units of expected revenue.

| Discount | Churn reduction | Expected saved revenue | Discount cost | Net revenue impact | ROI   |
|----------|-----------------|------------------------|---------------|--------------------|-------|
| 10%      | 3pp             | 109,968                | 45,840        | 64,128             | 1.40  |
| 10%      | 6pp             | 219,936                | 56,837        | 163,099            | 2.87  |
| 10%      | 10pp            | 366,561                | 71,499        | 295,061            | 4.13  |
| 20%      | 3pp             | 109,968                | 91,680        | 18,288             | 0.20  |
| 20%      | 6pp             | 219,936                | 113,674       | 106,262            | 0.93  |
| 20%      | 10pp            | 366,561                | 142,999       | 223,562            | 1.56  |
| 30%      | 3pp             | 109,968                | 137,521       | -27,552            | -0.20 |
| 30%      | 6pp             | 219,936                | 170,511       | 49,425             | 0.29  |
| 30%      | 10pp            | 366,561                | 214,498       | 152,062            | 0.71  |

This changes the recommendation. The model identifies who to consider, but the ROI analysis decides what offer is sensible. A 10% discount can be attractive even with modest churn reduction. A 30% discount needs a much stronger treatment effect and should not be rolled out broadly without testing.

## Retention Action Planner

The final user-level model output maps each user to a risk band and a recommended action.

Recommended operating logic:

- **Critical risk:** personalised save offer and customer-service contact for high-value users.

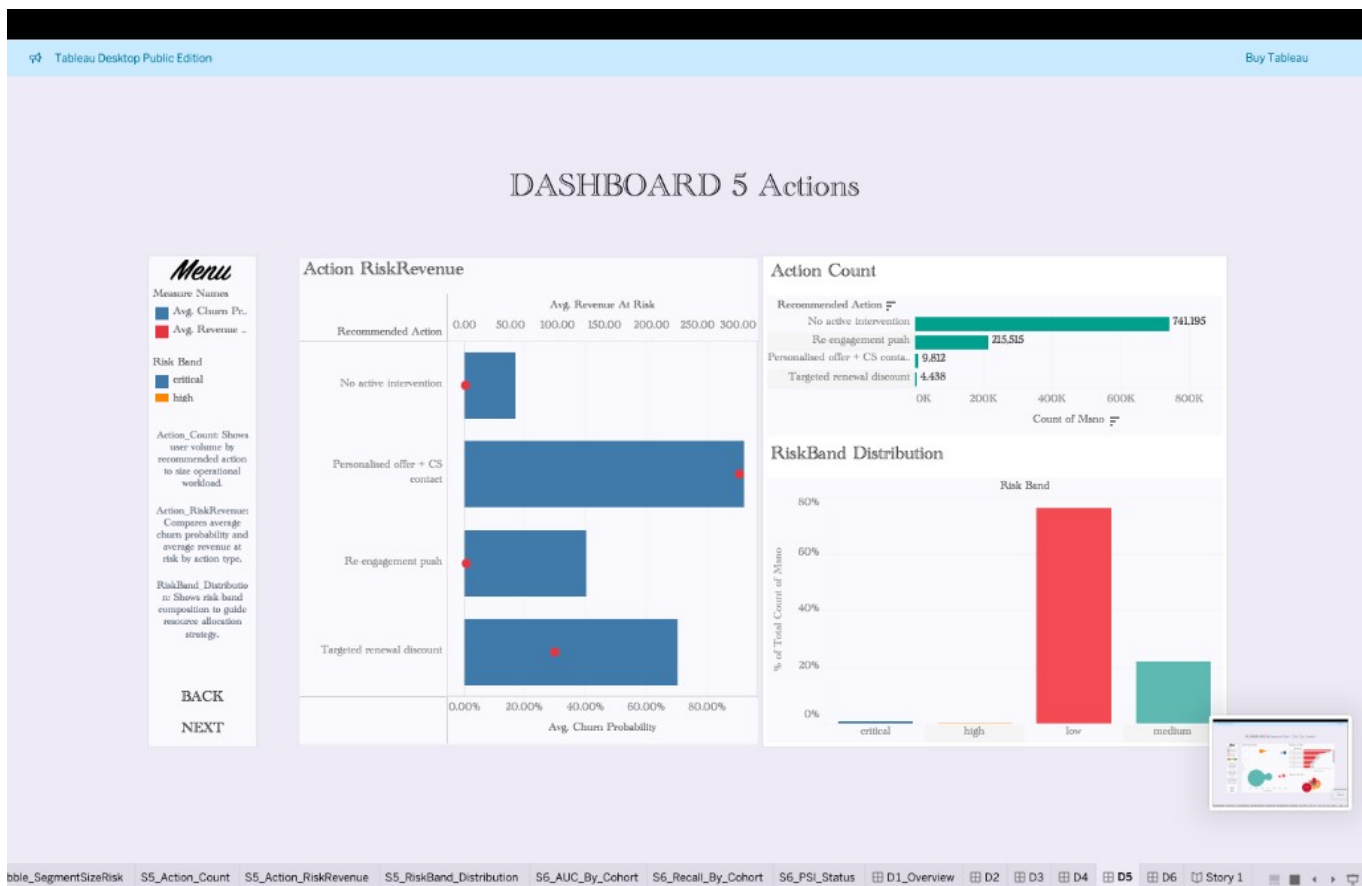


Figure 5: Retention action dashboard

- **High risk:** controlled renewal-incentive test.
- **Medium risk:** low-cost re-engagement messages and product nudges.
- **Low risk:** no paid intervention; continue normal lifecycle communication.

This prevents the business from wasting incentive budget on users who are unlikely to churn or users whose expected value does not justify expensive treatment.

## Dashboard Suite

The Tableau workbook is designed as a business-facing story, not just a collection of charts.



Figure 6: Executive overview dashboard

The overview dashboard summarises revenue, user volume, auto-renewal rate, cancellation rate, and monthly trend movement.

The churn drivers dashboard translates statistical outputs into business-readable evidence around auto-renewal, acquisition channel, discounting, and user profile patterns.

The monitoring dashboard tracks AUC, top-decile recall, and PSI across cohorts. This shows whether model performance and score distributions remain stable after deployment.

## Experiment Design

The recommended next step is not immediate full rollout. It is a controlled A/B test.

| Design choice | Recommendation                                    |
|---------------|---------------------------------------------------|
| Target users  | High and critical risk users approaching expiry   |
| Treatment     | Renewal incentive through email and in-app banner |
| Control       | Standard renewal reminder                         |



| Design choice  | Recommendation                                                          |
|----------------|-------------------------------------------------------------------------|
| Randomisation  | User-level 50/50 split                                                  |
| Primary metric | 30-day post-expiry retention                                            |
| Guardrails     | ARPU, complaint rate, auto-renew rate, net revenue after incentive cost |
| Decision rule  | Roll out only if retention uplift and net revenue are both positive     |

This connects model output to a measurable business decision. The model is useful only if the action improves retention without damaging net revenue.

## Skills Demonstrated

This project demonstrates:

- **Analytics engineering:** DuckDB warehouse, dbt models, source-to-mart lineage, repeatable SQL.
- **Data analysis:** churn KPIs, revenue exposure, segment prioritisation, dashboard-ready outputs.
- **Statistical thinking:** hypothesis testing, p-values, effect sizes, evidence-based claims.
- **Machine learning:** baseline model, LightGBM, calibration, lift-based evaluation, risk scoring.
- **Business judgement:** discount economics, ROI sensitivity, action prioritisation, A/B test design.
- **BI communication:** Tableau story dashboards and executive-friendly reporting.

## Final Recommendation

KKBox should use the model as a targeting and prioritisation layer, not as an automatic discount engine. The strongest business use case is to focus retention review on critical and high-risk users, test smaller incentives first, and scale only when the experiment proves positive net revenue.

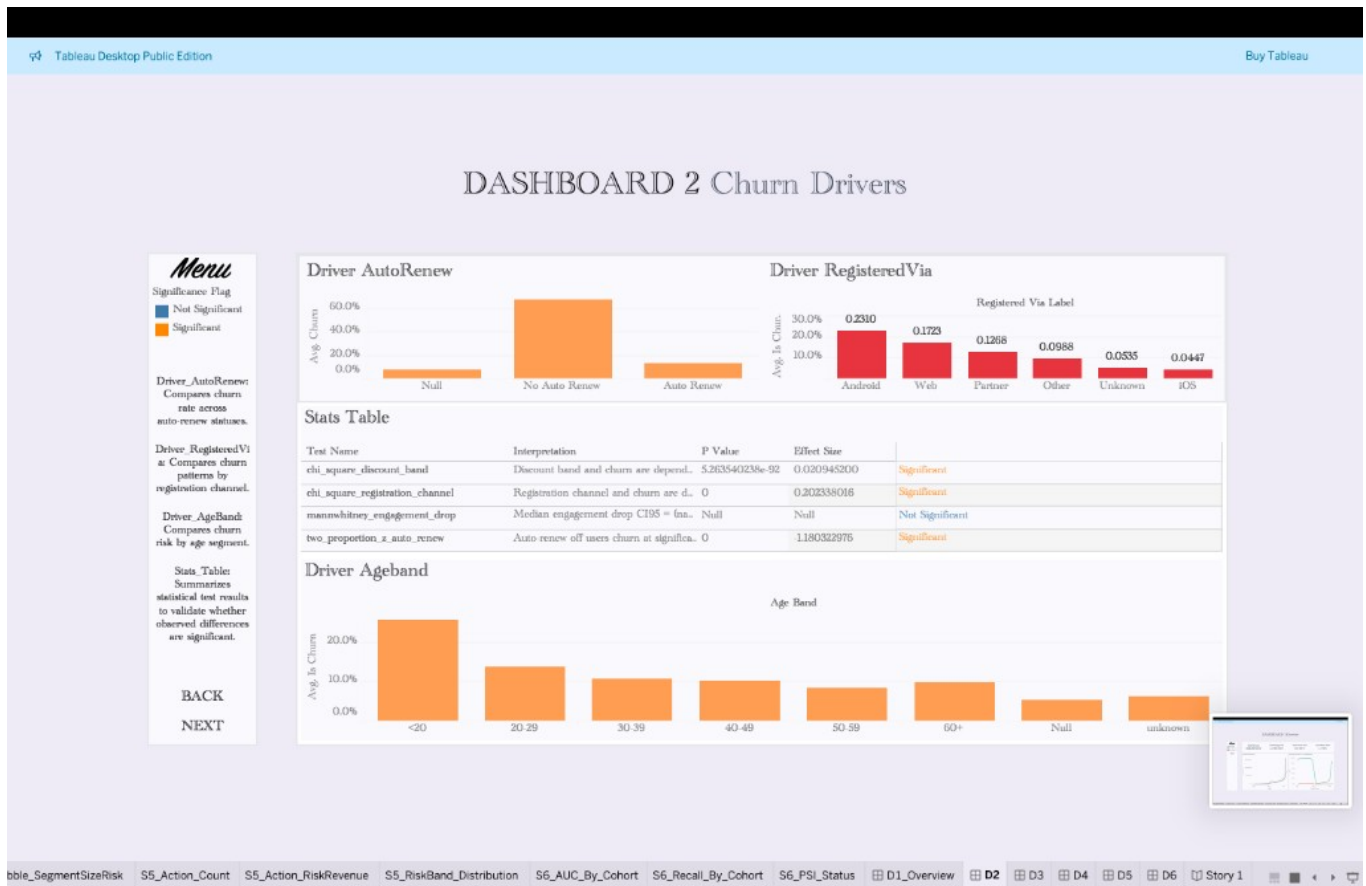


Figure 7: Churn drivers dashboard

